

Elvis Sierra

Software & Data Engineer // Systems, Pipelines & Backend Platforms

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SUMMARY

Engineer with **6+ years** building and operating the systems a business decision has to stand on: ETL pipelines, search and backend services, internal tooling, and the analytics layers on top of them. I take an ambiguous problem and give it clear edges, deciding what to build, naming the tradeoffs out loud, and owning the result from schema to shipped output. I have carried work **zero-to-one inside startups** and hardened it inside **larger, structured organizations**, which is why the title on the door has changed while the discipline underneath it has not. What an organization gets is an engineer who treats correctness as a contract, communicates in writing that outlives the conversation, and closes the loop between what a system does and what a decision needs.

CORE COMPETENCIES

Languages	Python, SQL, JavaScript
Data & Pipelines	ETL/ELT, Pandas, data validation and quality enforcement, data contracts, schema design, reporting automation, KPI and metric modeling, root cause analysis
Backend & APIs	Django, Django REST Framework, FastAPI, Flask, GraphQL, REST design, token-based auth and RBAC, service-oriented architecture
Data Stores	PostgreSQL, MySQL, OpenSearch, Redis, AWS RDS, S3
Cloud & DevOps	AWS (ECS, EC2, Lambda, API Gateway, CloudWatch), Docker, Airflow, Git-based CI/CD, Postman
Frontend	Vue.js, HTML/CSS, JavaScript SPA work
Practice	Zero-to-one system design, technical roadmapping, API and design specs, cross-functional KPI definition, self-audit against source
Learning	<i>Snowflake, dbt, Redshift, Kubernetes, Kafka</i>

Quality Analyst // Data Engineering & Reporting Automation

Apple Maps (Contract), Austin, TX

Apr 2025 – Present

- Own **ETL pipelines** that ingest file-based and relational inputs, run Python and Pandas transformations, and load structured tables built for recurring reporting and ad hoc analysis.
- Consolidated fragmented scripts into a **modular Python ETL package** with defined entry points, schema validation, quality enforcement, and stage-level logging, so a failure traces to the exact stage that caused it.
- Designed LLM-assisted workflows for semi-structured reporting inputs, then built the post-processing layer that validates and corrects those outputs against the relational data model. The model is the check, not the model output.
- Automated recurring reporting with **Airflow-scheduled jobs** that query analytical tables, compute aggregates, and populate presentation templates, eliminating **16 hours per week** of manual reporting.
- Partnered with reporting and operations stakeholders to define KPIs, calculation logic, and refresh cadence, so pipeline output matches the logic the decision is actually reasoned with.
- Ran exploratory analysis and data profiling to trace metric discrepancies to root cause before they reached downstream reports.

Tech: Python, SQL, Pandas, Airflow, PostgreSQL, AWS, Docker, Selenium

Software Engineer // Full-Stack & Backend

Media Co-Lab (Freelance), Austin, TX

Mar 2024 – Jan 2025

- Designed a **multi-tenant permission model** for a collaborative media platform, layering org and team approval gates, scoped ORM query paths, and object-level ownership checks so isolation held at three independent points instead of one filter a future change could quietly break.
- Chose server-side revocable tokens over stateless JWTs so an admin revoking a member's access took effect immediately. For a B2B tool, access control has to be instant.
- Designed normalized relational schemas with the right keys, indexes, and constraints, maintained through migrations as the model evolved.
- Cut response times on heavy read paths by profiling query plans and adding targeted single and composite indexes, reducing load on the database.
- Ran a self-audit of the codebase against its own documentation before a technical review, found a live bug in the uploads endpoint, fixed it, and adopted "verify against source, not docs" as a standing habit.

Tech: Python, Django, DRF, JavaScript, Vue.js, PostgreSQL, Redis, Docker, GraphQL

Senior Backend Engineer // Search & Backend Systems

ThinkOnward (Startup), Austin, TX

Sep 2022 – Feb 2024

- Built an **async, event-driven ingestion pipeline** where Lambda parsed S3 files, extracted structured fields, and indexed **100K documents into OpenSearch**, with a field-mapping redesign that prevented drift and preserved lineage.
- Cut query time on core reporting endpoints from **8 seconds to under 1 second** by redesigning index mappings and query shapes.
- Led **zero-to-one system design** from greenfield, turning ambiguous product goals into technical roadmaps with real boundaries and holding engineering, product, and customer-facing teams aligned as the architecture moved.
- Built a geospatial polygon system tied to the OpenSearch index for well-location queries.
- Wrote the backend API and design specs covering data flows, contracts, and failure modes for teams who were not in the room when the decisions were made.
- Ran containerized services on AWS with Git-based CI/CD (builds, schema migrations, automated testing), keeping rapid iteration from costing system stability.

Tech: Python, Django, PostgreSQL, OpenSearch, AWS (ECS, EC2, Lambda, S3), Docker, GraphQL, Airflow

Independent Contractor // Backend & Data Engineering

Freelance Software Development, Austin, TX

Aug 2020 – Sep 2022

- Delivered backend and data engineering for small clients and early-stage products, building internal tooling and services in Python, Django, and AWS.

Tech: Python, Django, AWS, PostgreSQL

Software Engineer // Performance & Analytics Systems

Caterpillar Inc., Victoria, TX

Mar 2018 – Aug 2020

- Designed and built a **REST API-backed performance-management system** that introduced traceability to factory workflows, ingesting raw data points once and exposing them as daily and weekly aggregates and trend views.
- Wrote the SQL transformation layer behind throughput, quality, and cost metrics, feeding the charts and dashboards operations leadership used to allocate time, work, and resources.
- Ran root cause analysis through KPI decomposition, trend analysis, and anomaly detection, then recommended workflow changes and measured the effect: **65% improvement in measured output quality** and a **50% cut in rework resource usage**.
- Defined requirements with QA and environmental teams so the system measured what the operation actually needed.

Tech: Python, Flask, SQL, MySQL, HTML/CSS, AWS

SELECTED PROJECTS

ReportingAuto Python, Pandas, ETL

A recipe-driven reporting engine. Column rules and analyses (correlations, crosstabs, outliers, time series) are configured once; every report after that runs itself, each output tagged with a run ID so it traces back to the recipe that produced it.

Logger FastAPI, Vue.js, PostgreSQL, Docker

A full-stack time tracker whose real work is the auth layer: HttpOnly-cookie sessions, refresh tokens that rotate on every use with reuse detection per RFC 6819, and a single-flight guard so a burst of concurrent requests cannot log a valid session out.

ShopFloor FastAPI, GraphQL, PostgreSQL, Vue.js

A manufacturing execution and quality-tracking system, work orders, routings, bills of materials, and a click-to-draw floor-plan editor over an SVG plant map, sitting behind one typed GraphQL API over 16 SQLAlchemy models with a deliberately flat, N+1-free schema.

Media Co-Lab Django, DRF, Vue.js, PostgreSQL, Docker

A multi-tenant media collaboration platform where organizations contain teams, every item carries a comment feed and a live WebSocket relay, and tenant isolation is held by a three-layer permission model rather than a single query filter.

EDUCATION

University of Houston–Victoria

Associate of Applied Science in Engineering